Solar Energy Forecasting System

Project Report

By
Rahul Gorana - AF05109277
Maximus Shaj Paul - AF05093472

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Acknowledgement

The project **"Solar Energy Forecasting System"** is the Project work carried out by

Name	Enrollment No
Rahul Gorana	**AF05109277**
Maximus Shaj Paul	**AF05093472**

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Abstract

Solar energy is one of the fastest-growing renewable energy sources and plays a vital role in reducing dependence on fossil fuels. However, solar power generation is highly dependent on environmental factors such as sunlight intensity and temperature, which makes it unpredictable. This uncertainty creates challenges in energy planning, efficient utilization, and maintaining grid stability.

The main objective of this project is to develop a system that can accurately forecast solar power generation using historical data and weather conditions. Such predictions help energy providers optimize power distribution, reduce losses, and improve overall efficiency in solar energy management.

To solve this problem, the Solar Energy Forecasting System is developed by integrating backend technologies and machine learning techniques. The system processes historical solar generation data and weather data from CSV files and external APIs. It uses Spring Boot for backend operations, MySQL for data storage, and the WEKA machine learning library to train a predictive model. Based on input features such as temperature and solar irradiation, the model generates future energy predictions. A dashboard is also provided to visualize historical trends and forecasted results.

This project provides a practical and efficient solution for improving solar energy forecasting and supports better decision-making in renewable energy systems.

1. Introduction

Solar energy has emerged as a reliable and sustainable source of power in modern energy systems. The increasing adoption of solar power plants has created a need for efficient monitoring and prediction of energy generation. Since solar energy production depends heavily on environmental conditions such as sunlight intensity and temperature, accurate forecasting becomes essential for proper energy management. This project focuses on developing a Solar Energy Forecasting System that utilizes historical generation data and weather information to predict future solar power output using machine learning techniques.

1. Background of the Study

The growing demand for clean and renewable energy has led to the widespread deployment of solar power plants. However, solar energy generation is inherently variable due to changing weather conditions. Traditional methods of estimation are not sufficient to handle this variability effectively. Advances in data analytics and machine learning have enabled more accurate forecasting by analyzing historical patterns and environmental factors, making it possible to improve the reliability of solar power systems.

2. Motivation

The main motivation behind this project is to address the challenge of unpredictable solar energy generation. Accurate forecasting helps in better planning of energy usage, reduces wastage, and improves grid stability. Additionally, the availability of real-world datasets and machine learning tools provides an opportunity to build a practical system that can deliver meaningful predictions and insights.

3. Significance of the Study

This project is significant as it contributes to improving the efficiency and reliability of solar energy systems. By providing accurate forecasts, it helps energy providers and industries make informed decisions regarding power distribution and utilization. It also supports the broader goal of promoting renewable energy adoption by enhancing its dependability and performance.

4. Features of Solar Energy Forecasting System

- Integration of historical solar generation and weather data
- Data ingestion from CSV files and external weather APIs
- Machine learning–based prediction using WEKA
- Storage and management of data using MySQL
- Backend processing using Spring Boot
- Visualization of historical and forecasted data through a dashboard
- Scalable and modular system architecture

5. Reporting and Analytics

The system provides reporting and analytical capabilities through a dashboard interface. Users can view historical generation trends, weather patterns, and predicted energy outputs in a structured format. These insights help in understanding system performance and enable better decision-making. Analytical data can be used for performance evaluation, forecasting accuracy assessment, and future planning.
to present data in a clear and understandable format.

2. System Analysis

The system analysis phase focuses on understanding the problem, defining system requirements, and evaluating the feasibility of developing the Solar Energy Forecasting System. This phase ensures that the proposed solution is practical, efficient, and meets user needs. It involves identifying system objectives, analyzing requirements, and designing a structured approach for implementation.

1. Problem Definition

Solar energy generation is highly dependent on weather conditions such as sunlight intensity and temperature, making it unpredictable. This variability creates challenges in efficient energy planning, resource allocation, and maintaining grid stability. Existing methods lack accuracy and real-time adaptability, leading to inefficiencies in solar energy utilization. Therefore, a system is required to accurately forecast solar power generation using data-driven techniques.

2. Objectives of the System

- To develop a system for predicting solar energy generation
- To utilize historical and weather data for accurate forecasting
- To implement machine learning techniques for prediction
- To store and manage large datasets efficiently
- To provide visualization of historical and predicted data
- To support better decision-making in energy management

3. Feasibility Study

a. Technical Feasibility
The system is technically feasible as it uses widely available technologies such as Spring Boot, MySQL, and WEKA. The required tools and frameworks are stable, well-documented, and suitable for building scalable applications.

b. Economic Feasibility
The project is cost-effective since it uses open-source technologies and requires minimal hardware resources. No expensive infrastructure or licensing costs are involved.

c. Operational Feasibility
The system is user-friendly and can be easily operated by users with basic technical knowledge. The dashboard and reporting features make it simple to understand and use.

d. Time Feasibility
The project can be completed within the given timeframe as the technologies used are efficient and development can be done in phases.

4. System Requirements

a. Functional Requirements

- Load and process CSV data (generation and weather data)
- Store data in the database
- Train machine learning model using historical data
- Predict future solar power generation
- Fetch weather data from external APIs
- Display data and predictions on dashboard

b. Non-Functional Requirements

- System should be scalable and efficient
- Fast data processing and response time
- Secure data handling
- Reliable and accurate predictions
- Maintainable and modular code structure

c. Hardware Requirements

- Minimum 4 GB RAM
- Processor: Intel i3 or above
- Storage: 10 GB free space
- System: Laptop/Desktop

d. Software Requirements

- Java (JDK 17 or above)
- Spring Boot Framework
- MySQL Database
- WEKA Machine Learning Library
- IDE (IntelliJ / Eclipse / VS Code)
- Web Browser

5. Proposed System

The proposed system is a Solar Energy Forecasting System that integrates data processing, machine learning, and visualization to predict solar power generation. It collects historical generation data and weather data from CSV files and external APIs, and stores them in a structured database using MySQL.

The system processes this data to generate training features, which are used by a machine learning model built using the WEKA library. Based on input parameters such as temperature and solar irradiation, the model predicts future solar energy output.

The system also provides a dashboard to visualize historical data and forecast results, enabling better analysis and decision-making. Overall, the system offers an efficient and scalable solution for accurate solar energy forecasting.

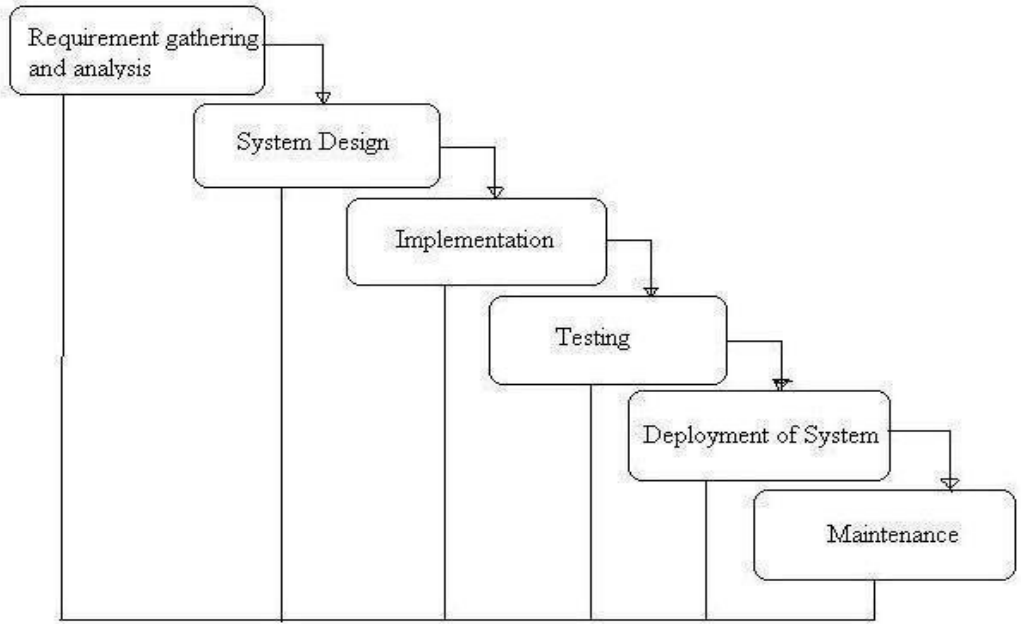


Fig. 5.1 . Phases of Development

6. Phases of Development

1. Requirement Analysis & System Study
In this phase, the problem of solar energy unpredictability is studied in detail. System requirements are gathered, including data sources, prediction needs, and user expectations. Existing methods are analyzed to identify gaps and limitations.

2. System Design
This phase involves designing the overall architecture of the system, including database schema, entity relationships, and system modules. Data flow between components such as data ingestion, processing, and prediction is clearly defined. Proper design ensures scalability and maintainability.

3. Implementation (Coding)
In this phase, the system is developed using Spring Boot for backend services. Database integration is implemented using JPA and MySQL, and CSV data ingestion is handled using OpenCSV. Machine learning functionality is implemented using the WEKA library.

4. Testing & Debugging
The system is thoroughly tested to ensure correct functionality. Unit testing and integration testing are performed to validate data processing, prediction accuracy, and system performance. Errors and bugs are identified and resolved.

5. Deployment & Maintenance
After successful testing, the system is deployed on a local or server environment. Regular maintenance is performed to update the model, fix issues, and improve system performance. Future updates may include new features and enhancements.

7. ER Diagram

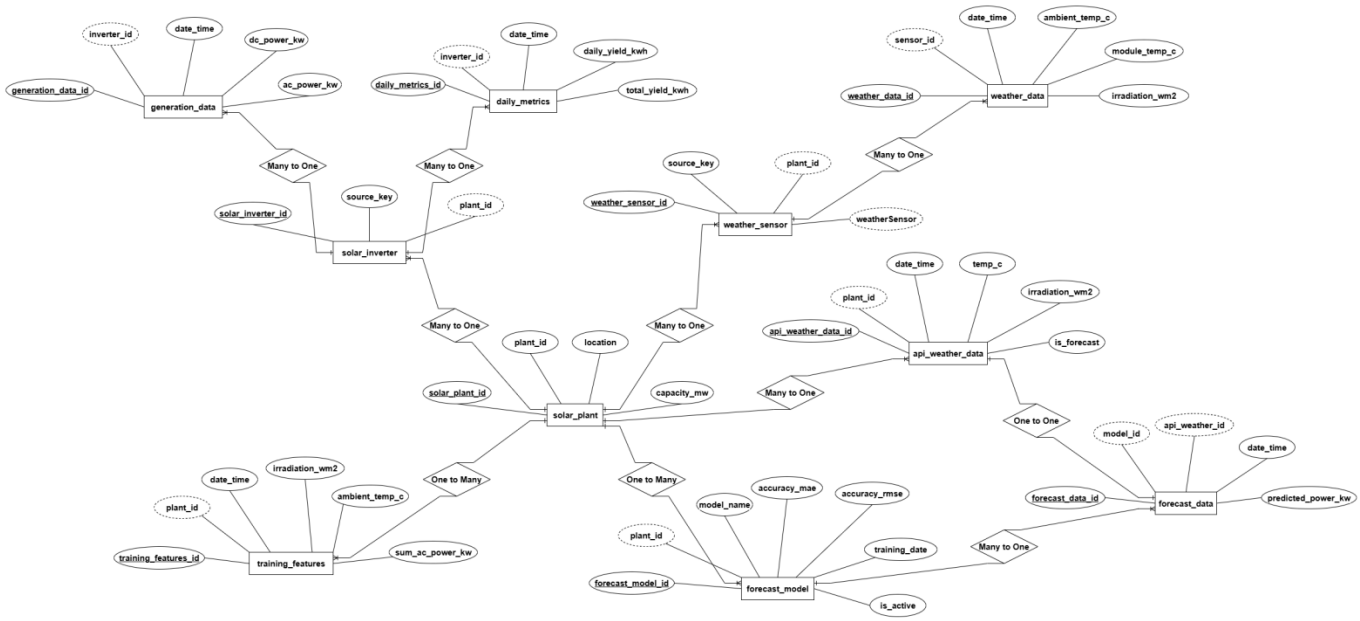


Fig. 7.1 . ER Diagram

The ER (Entity-Relationship) diagram represents the logical structure of the database used in the Solar Energy Forecasting System. It illustrates how different entities such as solar plant components, generation data, weather data, and forecasting models are interconnected. The diagram helps in understanding how data is stored, managed, and used for generating accurate predictions.

1. Entities and Their Attributes

- **SolarPlant**
  Attributes: plantId, location, capacityMw
  Description: Represents the main solar power plant. It acts as the root entity that connects all other components such as inverters, sensors, models, and data.
- **SolarInverter**
  Attributes: id, sourceKey
  Description: Represents the inverters installed in the plant. Each inverter converts DC power to AC power and generates power data.
- **GenerationData**
  Attributes: dateTime, dcPowerKw, acPowerKw
  Description: Stores time-based power generation data for each inverter, including both DC and AC power readings.
- **DailyMetrics**
  Attributes: dateTime, dailyYieldKwh, totalYieldKwh
  Description: Stores daily aggregated energy production values for each inverter. It helps in tracking daily performance and total lifetime energy generation.
- **WeatherSensor**
  Attributes: id, sourceKey
  Description: Represents physical weather sensors installed at the plant location to monitor environmental conditions.
- **WeatherData**
  Attributes: dateTime, ambientTempC, moduleTempC, irradiationWm2
  Description: Stores real-time weather data collected from sensors, which is essential for analyzing environmental impact on power generation.
- **ApiWeatherData**
  Attributes: dateTime, tempC, irradiationWm2, isForecast
  Description: Stores weather data fetched from external APIs. It includes both current and forecasted weather information used for prediction.
- **TrainingFeature**
  Attributes: dateTime, irradiationWm2, ambientTempC, sumAcPowerKw
  Description: Represents processed data derived from generation and weather datasets. It is used as input for training the machine learning model.

- **ForecastModel**
  Attributes: modelName, accuracyMae, accuracyRmse, trainingDate, isActive
  Description: Stores information about trained machine learning models, including their accuracy and status.
- **ForecastData**
  Attributes: dateTime, predictedPowerKw
  Description: Stores predicted solar energy output generated by the machine learning model.

2. Relationships

- A SolarPlant has a one-to-many relationship with SolarInverter, meaning one plant contains multiple inverters.
- A SolarInverter has a one-to-many relationship with GenerationData, as each inverter produces continuous generation data over time.
- A SolarInverter has a one-to-many relationship with DailyMetrics, since daily performance metrics are recorded for each inverter.
- A SolarPlant has a one-to-many relationship with WeatherSensor, as multiple sensors can be installed in a plant.
- A WeatherSensor has a one-to-many relationship with WeatherData, as it records environmental data periodically.
- A SolarPlant has a one-to-many relationship with ApiWeatherData, since multiple weather records (current and forecasted) are associated with the plant.
- GenerationData and WeatherData are combined to form TrainingFeature, representing a derived relationship used for machine learning.
- A SolarPlant has a one-to-many relationship with ForecastModel, as multiple models can be trained over time.
- A ForecastModel has a one-to-many relationship with ForecastData, as each model generates multiple predictions.
- Each ForecastData is linked to ApiWeatherData, ensuring predictions are based on corresponding weather inputs.

3. ER Diagram Summary

The ER diagram provides a structured representation of how data flows through the system, from raw data collection to prediction generation. It highlights the relationships between plant components, environmental data, and machine learning models, ensuring efficient data management and accurate forecasting.

3. System design

The system design phase focuses on defining the architecture, components, and data structures of the Solar Energy Forecasting System. It provides a detailed blueprint of how the system is organized, how different modules interact, and how data is structured and managed. The design ensures that the system is scalable, maintainable, and efficient in handling data processing and prediction tasks.

A) Modules

The system is divided into multiple modules, each responsible for a specific functionality:

1. Data Ingestion Module
This module is responsible for loading and importing data into the system. It reads historical solar generation data and weather data from CSV files and fetches real-time weather data from external APIs. The data is validated and prepared for storage.

2. Data Processing Module
This module processes raw data and converts it into structured formats. It cleans the data, handles missing values, and combines generation and weather data to create meaningful training features.

3. Database Management Module
This module manages all database operations such as storing, retrieving, updating, and deleting data. It ensures data consistency and efficient access using MySQL and JPA.

4. Machine Learning Module
This module is responsible for training and managing the predictive model using the WEKA library. It uses training features to build a regression model and evaluates its performance.

5. Prediction Module
This module uses the trained machine learning model to predict future solar energy generation based on input weather data.

6. API Integration Module
This module connects the system with external weather APIs to fetch real-time and forecasted weather data required for prediction.

7. Visualization & Dashboard Module
This module provides a user interface to display historical data, trends, and predicted results in a graphical and easy-to-understand format.

B) Data Structure of All Modules

1. Customized Tables

The system uses several customized database tables to store and manage data efficiently:

- **SolarPlant Table**
  Stores information about solar plants including plant ID, location, and capacity.
- **SolarInverter Table**
  Stores details of inverters associated with each plant.
- **GenerationData Table**
  Stores time-based solar power generation data such as DC and AC power.
- **DailyMetrics Table**
  Stores daily aggregated energy production values such as daily yield and total yield.
- **WeatherSensor Table**
  Stores information about weather sensors installed in the plant.
- **WeatherData Table**
  Stores real-time environmental data such as temperature and irradiation.
- **ApiWeatherData Table**
  Stores weather data fetched from external APIs, including forecast data.
- **TrainingFeature Table**
  Stores processed data used for training the machine learning model.
- **ForecastModel Table**
  Stores details of trained machine learning models and their performance metrics.
- **ForecastData Table**
  Stores predicted solar power output values.

2. Relationships Between Tables (ER/Class Diagram)

The system database follows a relational structure where different tables are interconnected:

- A SolarPlant is linked to multiple SolarInverter records
- Each SolarInverter is associated with multiple GenerationData and DailyMetrics records
- A SolarPlant is connected to multiple WeatherSensor records
- Each WeatherSensor generates multiple WeatherData records
- ApiWeatherData is associated with the plant and used for prediction inputs
- GenerationData and WeatherData are combined to form TrainingFeature data
- A ForecastModel is linked to the plant and generates multiple ForecastData records
- ForecastData is linked with ApiWeatherData for prediction reference

This structured relationship ensures efficient data flow from data collection to prediction and visualization.

3. System Architecture Flow

The System Architecture Flow describes how data moves through different components of the Solar Energy Forecasting System, from data collection to prediction and visualization. The system follows a modular and layered architecture to ensure scalability, maintainability, and efficient processing.

1. **Data Collection Layer**
   The system collects data from two main sources:
   - Historical solar generation and weather data from CSV files
   - Real-time and forecasted weather data from external APIs
     This data serves as the foundation for training and prediction.

2. **Data Ingestion Layer**
   The collected data is loaded into the system using data ingestion services. CSV files are parsed, validated, and converted into structured formats. API data is fetched and stored for further processing.

3. **Data Storage Layer**
   The processed data is stored in a relational database (MySQL). Different tables such as GenerationData, WeatherData, ApiWeatherData, and DailyMetrics are used to organize and manage the data efficiently.

4. **Data Processing & Feature Engineering Layer**
   In this layer, raw data is cleaned and transformed. Generation data and weather data are combined to create training features such as temperature, irradiation, and power output. This step is crucial for improving model accuracy.

5. **Machine Learning Layer**
   The processed training data is used to train a machine learning model using the WEKA library. The model learns the relationship between weather conditions and solar power generation.

6. **Prediction Layer**
   The trained model takes input from forecasted weather data (ApiWeatherData) and predicts future solar energy generation. The results are stored in the ForecastData table.

7. **Visualization & Reporting Layer**
   The predicted results and historical data are displayed through a dashboard. Users can view trends, compare past and predicted values, and analyze system performance.

8. **User Interaction Layer**
   Users interact with the system through the dashboard interface, where they can access reports, visualize data, and monitor predictions.

C) Screenshots

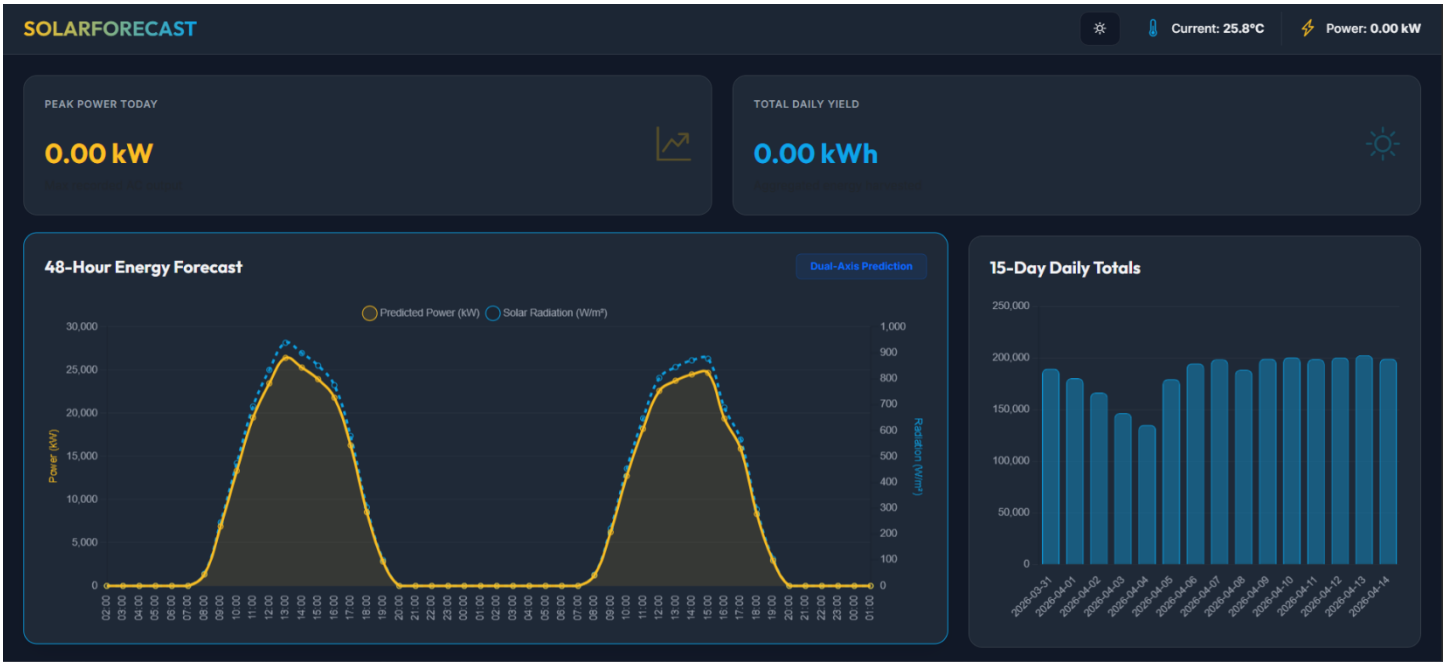
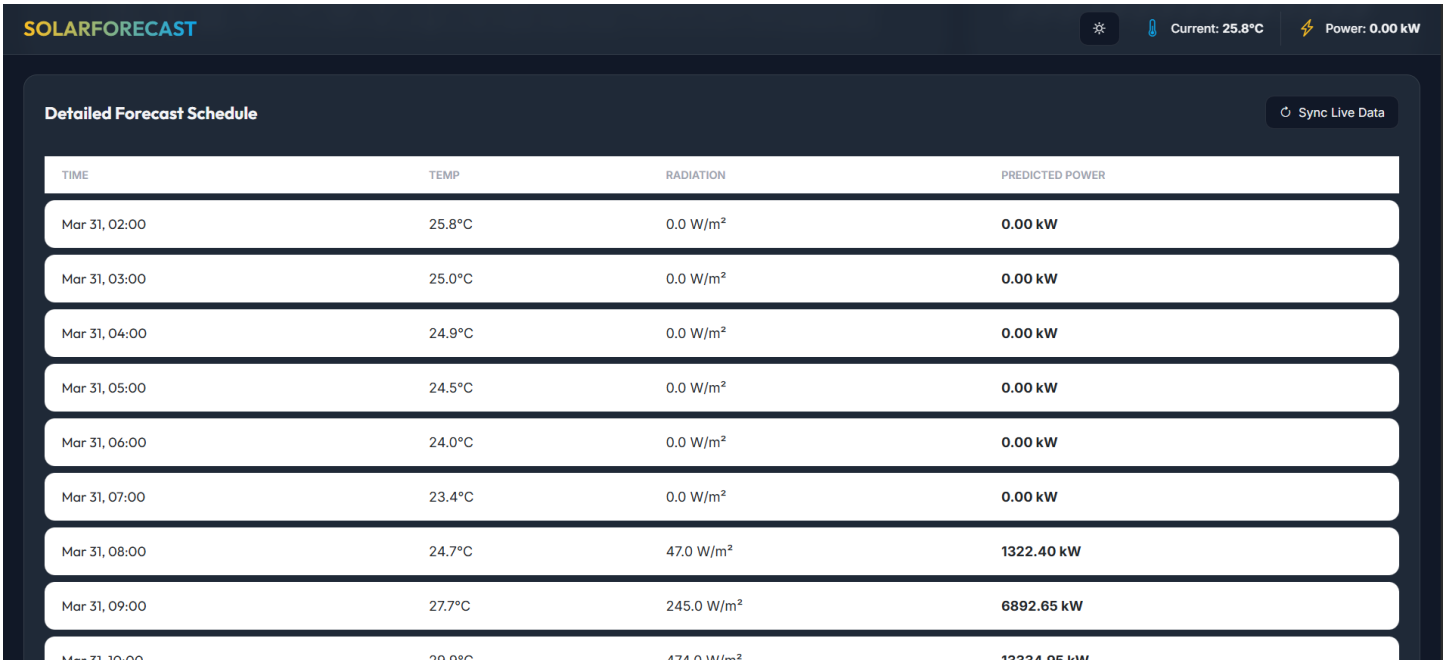


Fig. 3.1 . Live Data and Predicted Graphs



TIME	TEMP	RADIATION	PREDICTED POWER
Mar 31, 02:00	25.8°C	0.0 W/m²	0.00 kW
Mar 31, 03:00	25.0°C	0.0 W/m²	0.00 kW
Mar 31, 04:00	24.9°C	0.0 W/m²	0.00 kW
Mar 31, 05:00	24.5°C	0.0 W/m²	0.00 kW
Mar 31, 06:00	24.0°C	0.0 W/m²	0.00 kW
Mar 31, 07:00	23.4°C	0.0 W/m²	0.00 kW
Mar 31, 08:00	24.7°C	47.0 W/m²	1322.40 kW
Mar 31, 09:00	27.7°C	245.0 W/m²	6892.65 kW

Fig. 3.2 . Predicted Rows

4. Implementation

The implementation phase focuses on the actual development of the Solar Energy Forecasting System. It involves integrating various technologies and modules to build a fully functional system capable of data processing, prediction, and visualization. The system is developed using a combination of backend frameworks, database systems, and machine learning tools.

1. Technology Stack

The system is built using the following technologies:

- **Programming Language:** Java
- **Backend Framework:** Spring Boot
- **Database:** MySQL
- **Machine Learning Library:** WEKA
- **ORM Tool:** Spring Data JPA (Hibernate)
- **API Integration:** REST APIs for weather data
- **Data Handling:** CSV file processing (OpenCSV)
- **Development Tools:** IntelliJ IDEA / Eclipse / VS Code
- **Build Tool:** Maven

2. Backend Implementation

The backend of the system is developed using the Spring Boot framework. It follows a layered architecture consisting of controllers, services, and repositories. Controllers handle incoming requests, services contain business logic, and repositories manage database interactions using JPA.
The backend is responsible for data ingestion, processing, model training, prediction, and API integration. It also exposes REST endpoints for accessing data and predictions. The modular design ensures scalability, maintainability, and efficient handling of operations.

3. Database Implementation

The system uses MySQL as the relational database to store and manage data. Tables are designed based on entities such as SolarPlant, SolarInverter, GenerationData, DailyMetrics, WeatherSensor, WeatherData, ApiWeatherData, TrainingFeature, ForecastModel, and ForecastData.
Relationships between tables are maintained using foreign keys. JPA annotations are used to map Java classes to database tables. The database supports efficient storage, retrieval, and querying of large datasets required for forecasting.

4. Machine Learning Implementation

The machine learning component is implemented using the WEKA library. The system prepares training data by combining generation data and weather data into structured features. These features are used to train a regression model that predicts solar power output.
Model evaluation is performed using accuracy metrics such as MAE and RMSE. Once trained, the model is saved and reused for making predictions based on new weather data. This approach ensures efficient and accurate forecasting.

5. API Implementation

The system integrates external weather APIs to fetch real-time and forecasted weather data. REST API calls are used to retrieve data such as temperature and solar irradiation.
The fetched data is parsed, validated, and stored in the ApiWeatherData table. This data is then used as input for the prediction module. API integration ensures that the system can provide up-to-date and accurate forecasts.

6. Frontend Implementation

The frontend of the system is designed to provide a simple and user-friendly interface. It includes a dashboard that displays historical data, weather conditions, and predicted solar energy output.
Data is presented in the form of charts, tables, and graphs for better understanding. The frontend communicates with the backend through REST APIs to fetch and display data dynamically.

7. Security Implementation

Basic security measures are implemented to ensure safe access to the system. These include input validation, error handling, and controlled access to APIs.
Sensitive data is protected, and only authorized users can access system functionalities. The backend ensures secure communication and prevents unauthorized data manipulation.

5. Results and Discussion

The Results and Discussion section presents the outcomes of the implemented Solar Energy Forecasting System and evaluates its performance based on functional and non-functional aspects. It also discusses how well the system meets its objectives, along with challenges, limitations, and future improvements.

1. Results

- **Functional Results**

The system was successfully developed and tested, and all modules are functioning as expected:

  - **Data Ingestion Module**
    Successfully loads historical generation and weather data from CSV files and fetches real-time data from external APIs.
  - **Data Processing Module**
    Accurately cleans, processes, and combines raw data into structured training features required for model training.
  - **Database Management Module**
    Efficiently stores and retrieves large volumes of data using MySQL with proper relationships between tables.
  - **Machine Learning Module**
    Successfully trains a predictive model using the WEKA library and evaluates performance using accuracy metrics.
  - **Prediction Module**
    Generates future solar energy forecasts based on input weather data with reasonable accuracy.
  - **API Integration Module**
    Retrieves real-time and forecasted weather data reliably and integrates it into the prediction workflow.
  - **Visualization & Dashboard Module**
    Displays historical data and predicted results through charts and tables, enabling easy analysis.

- **Non-Functional Results**
  - **Usability**
    The system provides a simple and user-friendly interface, making it easy for users to understand and interact with data and predictions.
  - **Performance**
    The system processes data efficiently and generates predictions within acceptable response time.
  - **Reliability**
    The system performs consistently under normal conditions and produces stable prediction results.
  - **Data Integrity**
    Data is stored accurately with proper validation and relationships, ensuring consistency and correctness.

- **Scalability**
  The modular architecture allows the system to handle increasing data volume and future enhancements.

2. Discussion

1. Achievement of Objectives
The system successfully meets its objectives by accurately predicting solar energy generation using machine learning techniques. It integrates data processing, model training, and visualization into a single platform, fulfilling the intended goals of the project.

2. User Experience
The system provides a smooth and intuitive user experience through its dashboard interface. Users can easily view historical data, analyze trends, and understand prediction results without technical complexity.

3. Challenges Faced
- Handling missing or inconsistent data in CSV files
- Integrating multiple data sources (generation and weather data)
- Preparing accurate training features for machine learning
- Ensuring proper mapping between database entities
- Achieving good prediction accuracy with limited data

4. Limitations
- Prediction accuracy depends on the quality and quantity of data
- Limited to specific parameters like temperature and irradiation
- Does not include advanced deep learning models
- Real-time scalability is limited to current system design

5. Future Enhancements
- Integration of advanced machine learning or deep learning models
- Inclusion of more weather parameters (humidity, wind speed, etc.)
- Real-time data streaming and live prediction updates
- Improved dashboard with advanced analytics and visualizations
- Deployment on cloud platforms for better scalability

6. Conclusion

The Solar Energy Forecasting System successfully demonstrates the use of machine learning and modern backend technologies to predict solar power generation. By utilizing historical generation data and weather parameters such as temperature and solar irradiation, the system is able to generate reliable forecasts that support better energy planning and management.

The integration of Spring Boot, MySQL, and the WEKA machine learning library ensures efficient data processing, storage, and prediction. The modular architecture of the system allows smooth data flow from collection to visualization, making the system scalable and easy to maintain. The dashboard further enhances usability by providing clear insights into historical trends and predicted values.

Overall, the project achieves its objective of developing an effective and practical solar energy forecasting solution. It highlights the importance of data-driven approaches in renewable energy systems and provides a strong foundation for future improvements and real-world deployment.

7. Future scope

The Solar Energy Forecasting System can be further enhanced by incorporating advanced technologies and expanding its capabilities. Future improvements can focus on increasing prediction accuracy, improving real-time performance, and making the system more scalable and user-friendly. By integrating additional data sources and modern machine learning techniques, the system can evolve into a more robust and intelligent energy management solution.

Future Enhancements

- Integration of advanced machine learning and deep learning models to improve prediction accuracy
- Inclusion of additional weather parameters such as humidity, wind speed, and cloud cover
- Implementation of real-time data streaming and live prediction updates
- Deployment on cloud platforms for better scalability and accessibility
- Enhancement of dashboard with advanced analytics and interactive visualizations
- Integration with IoT-based sensors for automatic real-time data collection
- Support for multiple solar plants and large-scale energy systems
- Improvement in model performance through continuous learning and updates
- Development of a mobile application for easy monitoring and access
- Implementation of alert and notification systems for energy fluctuations

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